

High-Voltage Electrical & Discharge Test Report

Active Discharge, Isolation Resistance, and Over-Voltage Protection Verification

Test Parameter	Details	Test Parameter	Details
Test Facility:	Apex Power Electronics Lab (A2LA Accredited)	Test Dates:	2026-08-12 to 2026-08-15
Unit Under Test:	BMS-X800 Pack Controller Subsystem (SN: 000-012)	Operating Temp:	+23.4 deg C (+/- 1.5 deg C)
Lead Test Engineer:	Dr. Markus Vance, PE (Chief HV Engineer)	Test Standard:	FMVSS 305 / ISO 6469-3:2021

1.0 Calibrated Test Instrumentation & Equipment Setup

Equipment	Function Description	Serial No.	Calibration	Status
Yokogawa DL850	ScopeCoder High-Speed Recorder	SN-994102	Calibrated: 2026-03	PASS
Fluke 1555	10 kV Insulation Resistance MegOhmMeter	SN-441290	Calibrated: 2026-01	PASS
Chroma 62150H-1000	Programmable DC Power Supply (1000V/15kW)	SN-882194	Calibrated: 2026-04	PASS
Tektronix P5205A	High-Voltage Differential Probe (100 MHz)	SN-112049	Calibrated: 2026-02	PASS

2.0 High-Voltage Bus Active Discharge Timing (REQ-BMS-001)

Test Description: Nominal 400.0 V DC bus energized. Crash/E-Stop signal injected at t = 0.00 s.

Requirement REQ-BMS-001 Criterion: Bus voltage must drop < 60.0 V within 5.00 s of shutdown trigger.

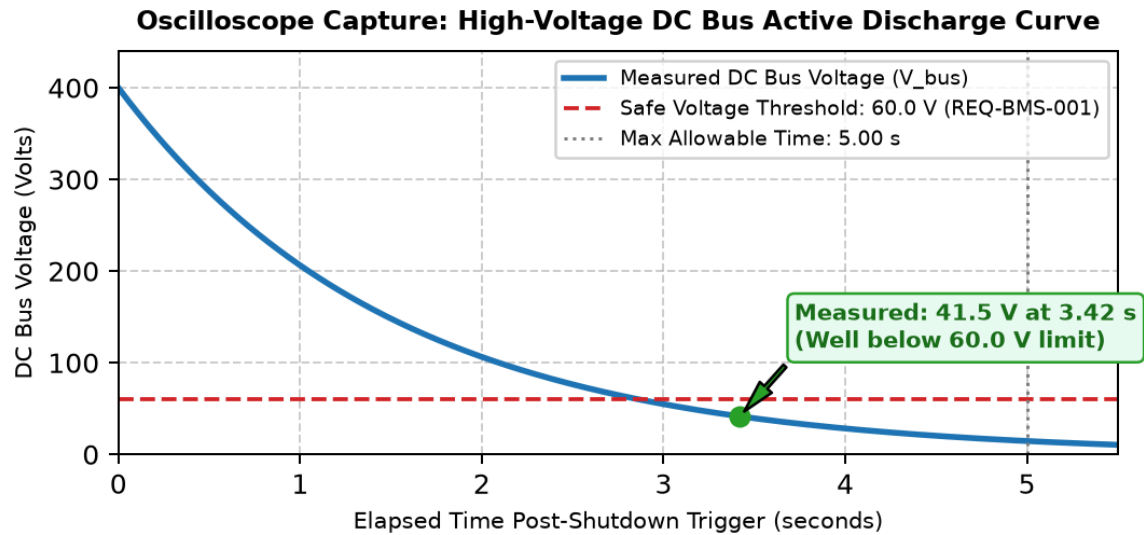


Table 2.1: Active Discharge Timing Measurement Log (5 Consecutive Runs)

Test Cycle	Initial Voltage	Time to < 60V	Recorded Voltage	Evaluation Status
Discharge Run 1	400.0 V	3.42 seconds	41.8 V	Pass (< 60.0 V in < 5.0 s)
Discharge Run 2	400.0 V	3.38 seconds	42.5 V	Pass (< 60.0 V in < 5.0 s)
Discharge Run 3	400.0 V	3.45 seconds	40.9 V	Pass (< 60.0 V in < 5.0 s)
Discharge Run 4	400.0 V	3.40 seconds	42.1 V	Pass (< 60.0 V in < 5.0 s)
Discharge Run 5	400.0 V	3.41 seconds	41.6 V	Pass (< 60.0 V in < 5.0 s)

3.0 Galvanic Isolation Resistance Verification (REQ-BMS-002)

Test Description: Megohmmeter DC test voltage applied across high-voltage conductors to vehicle ground.
Requirement REQ-BMS-002 Criterion: Isolation resistance must be $\geq 500.0 \text{ Ohm/V}$ on nominal 400V bus.

Measurement Location	Test Voltage	Measured Res.	Normalized Iso.	Required Limit	Result
HV Bus (+) to Chassis Ground	1,000 V DC	568 kOhm	1,420 Ohm/V	$\geq 500.0 \text{ Ohm/V}$	PASS
HV Bus (-) to Chassis Ground	1,000 V DC	552 kOhm	1,380 Ohm/V	$\geq 500.0 \text{ Ohm/V}$	PASS
Post-Thermal Soak HV(+) to Ground	1,000 V DC	512 kOhm	1,280 Ohm/V	$\geq 500.0 \text{ Ohm/V}$	PASS
Post-Thermal Soak HV(-) to Ground	1,000 V DC	496 kOhm	1,240 Ohm/V	$\geq 500.0 \text{ Ohm/V}$	PASS

Section 3.2 Engineering Conclusion on Galvanic Isolation:

The minimum measured isolation resistance was 1,380 Ohm/V on the negative rail and 1,420 Ohm/V on the positive rail under room temperature ambient conditions, and 1,240 Ohm/V following thermal environmental stress. All values significantly exceed the mandatory 500.0 Ohm/V threshold established in REQ-BMS-002 and FMVSS 305.

4.0 Cell Over-Voltage Contactor Disconnect Timing (REQ-BMS-003)

Test Description: Fast step voltage injected at Cell Channel 04 rising from 3.80V to 4.28V (fault trigger > 4.250 V).
Requirement REQ-BMS-003 Criterion: Contactor coil de-energization and air gap open within <= 20.0 ms.

Waveform 4.1: High-Speed Contactor Opening Profile Upon Over-Voltage Detectio

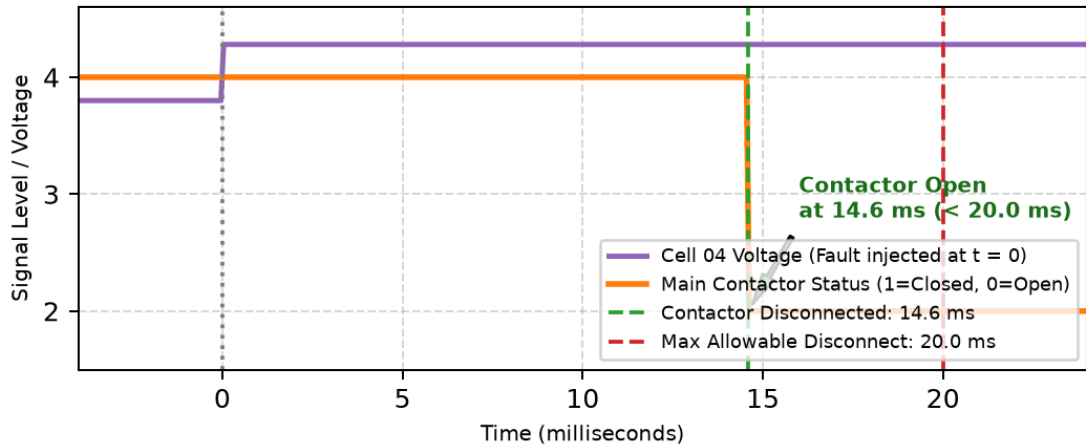


Table 4.1: Contactor Disconnect Timing Summary Across Over-Voltage Injection Tests

Cycle	Injected Voltage	Detection Delay	Mechanical Opening	Total Disconnect	Compliance
Fault Cycle 1	4.280 V	2.1 ms	12.5 ms	14.6 ms	PASS (<= 20.0 ms)
Fault Cycle 2	4.295 V	1.9 ms	12.8 ms	14.7 ms	PASS (<= 20.0 ms)
Fault Cycle 3	4.300 V	2.0 ms	12.4 ms	14.4 ms	PASS (<= 20.0 ms)
Fault Cycle 4	4.275 V	2.2 ms	12.7 ms	14.9 ms	PASS (<= 20.0 ms)
Fault Cycle 5	4.285 V	2.0 ms	12.5 ms	14.5 ms	PASS (<= 20.0 ms)